Reason. Plan. Act. Physical AI Workshop

Lecture 4b: Day 4 Recap

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Recap

Learning from Data

A central goal of machine learning is to automatically discover patterns from data. Instead of manually programming rules, we provide the system with data and allow it to learn a model.

Examples of tasks that can be learned from data:

- Recognizing objects in images
- Predicting house prices
- Detecting anomalies in sensor readings
- Classifying medical images



Different learning paradigms exist depending on the type of data and feedback available.

Supervised Learning Setup

In practice, we split the available dataset into three parts:

Training set S_{train}:

- used to **learn the model parameters**
- optimize empirical risk:

Validation (evaluation) set S_{val}:

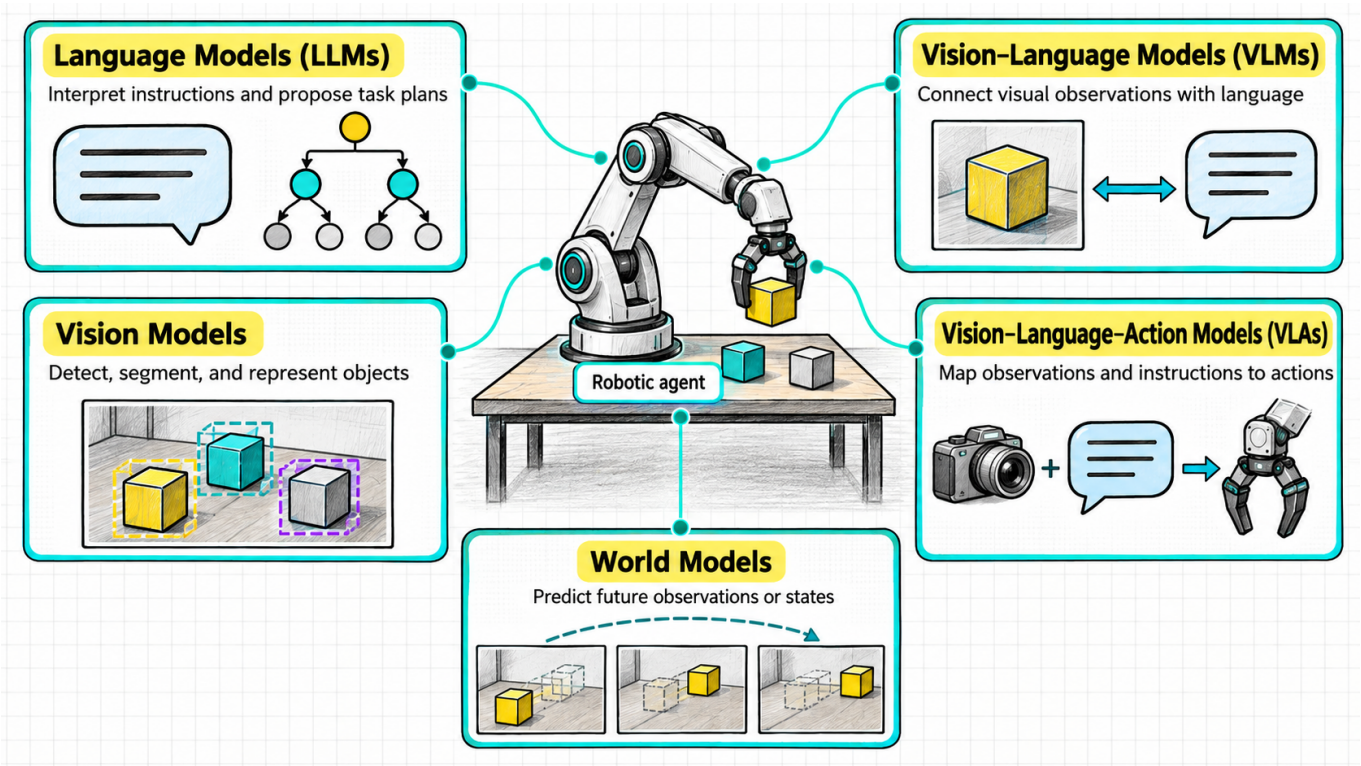
- used to **tune hyperparameters** (e.g., model size)
- helps detect **overfitting**

Test set S_{test}:

- used **only once at the end**
- provides an **unbiased estimate of generalization performance**

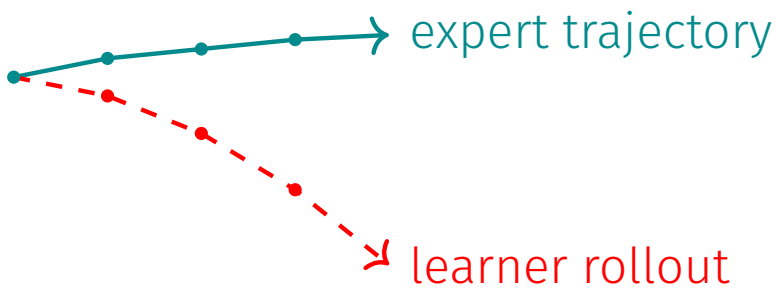
Never use test data during training or model selection.

Foundation Models for Robotics



Challenges

Limitations of Behavioral Cloning



Distribution Shift: The agent follows π_θ trained on (s, a) from expert demonstrations, but encounters states that were never demonstrated.

Key challenge: Errors compound over time.

- Small prediction error leads to a slightly wrong action
- The system transitions to a new, unseen state
- Errors increase as the agent moves further from expert behavior

Effect: performance degrades rapidly.

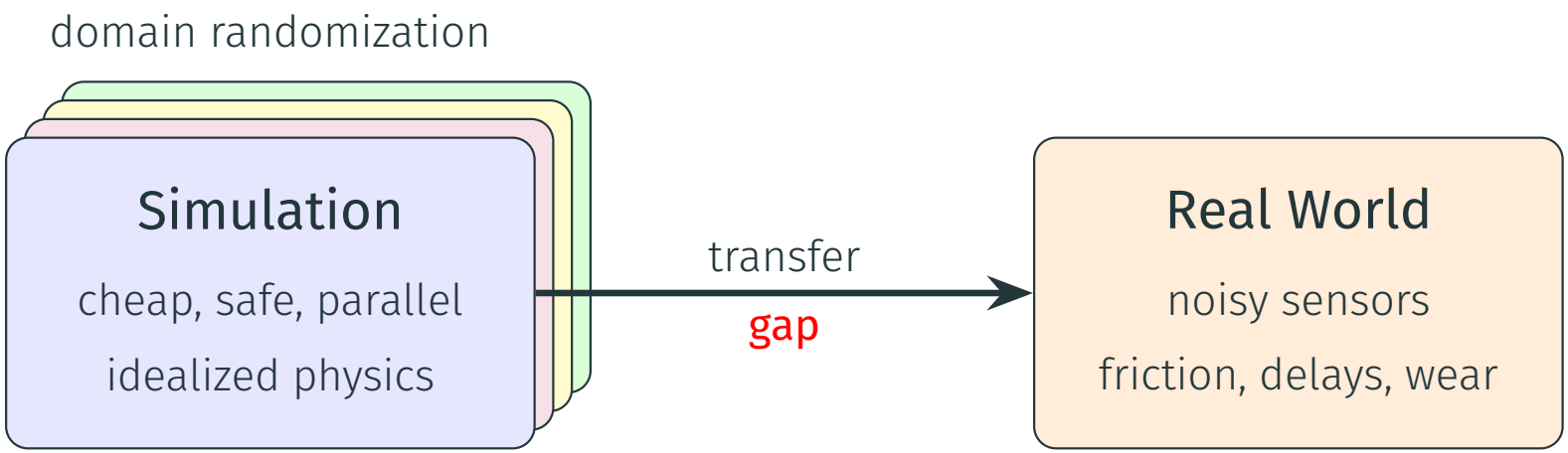
Limitations of Behavioral Cloning





Is this the same observation?

Sim-to-Real Gap

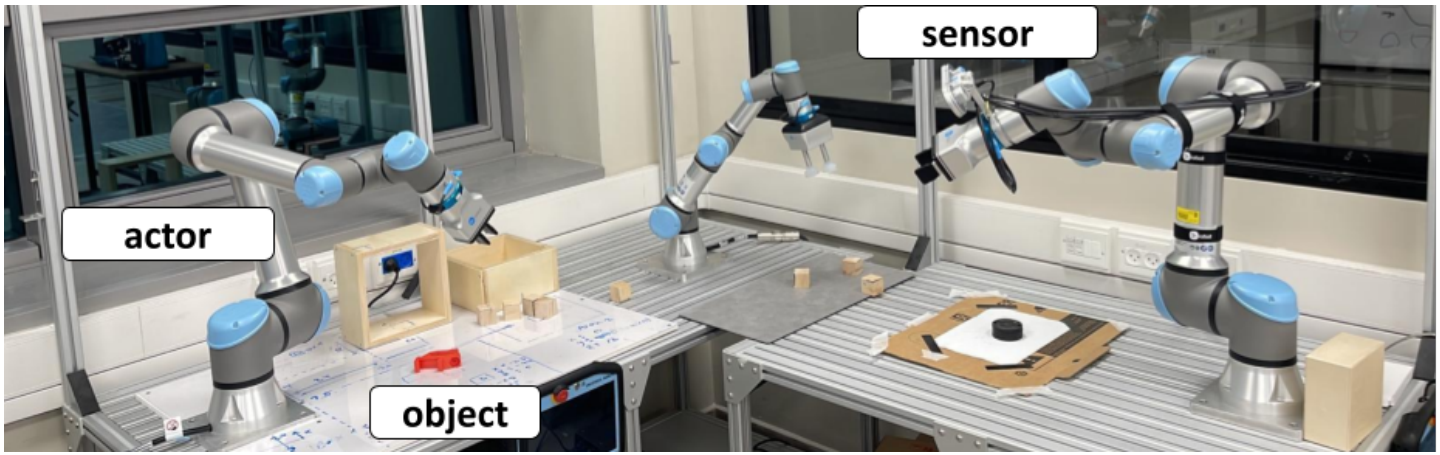


Causes: dynamics mismatch, rendered vs. real images, unmodeled effects (latency, wear)

Fixes: domain randomization, system identification, domain adaptation, real-world fine-tuning

CLAIR lab Examples

From Beliefs to Decisions



Value of Assistance for Grasping

Mohammad Masarwy, Yuval Goshen, David Dovrat, Sarah Keren

`https://arxiv.org/abs/2310.14402`

Belief Update

When receiving an observation o, the agent updates its current belief β using its **belief update function** $\tau : \mathcal{B} \times \Omega \times \mathcal{S} \mapsto \mathcal{B}$ which maps belief $\beta \in \mathcal{B}$ to the new belief.

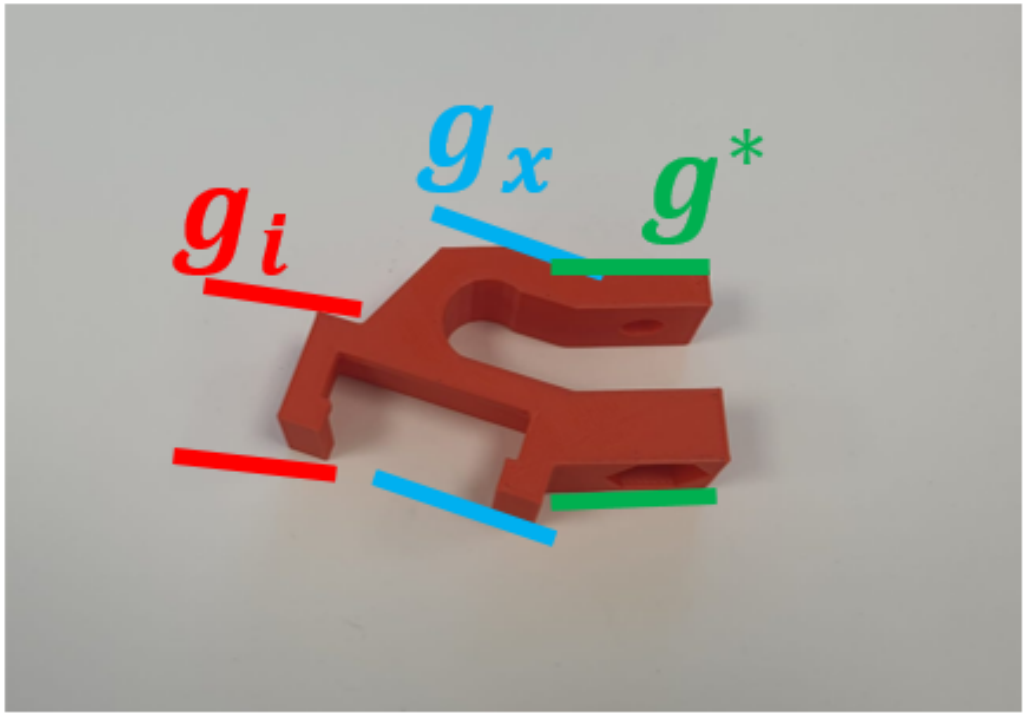
Commonly, a **Bayesian filter** is used:

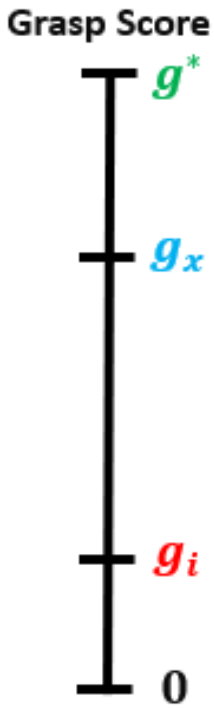
$$\beta^{o,a} = \frac{\mathcal{P}\left(o|s,a\right)\beta\left(s\right)}{\int_{s' \in \mathcal{S}} \mathcal{P}\left(o|s',a\right)\beta\left(s'\right)ds'} \tag{1}$$

where $\mathcal{P}$ is the sensor function and $\beta\left(s\right)$ is the estimated probability that s is the actual world state
Discrete version:

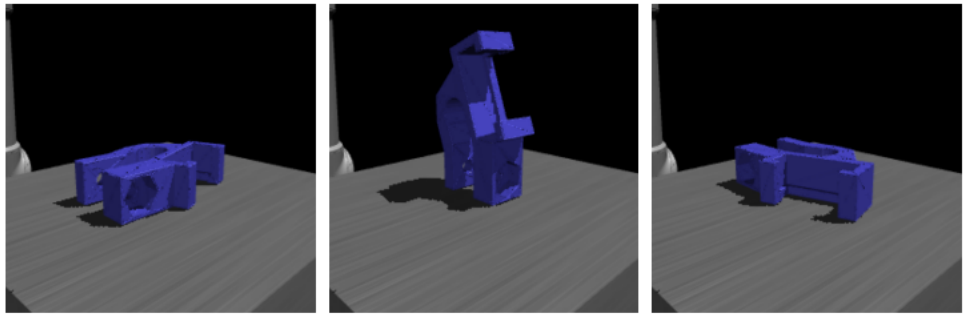
$$\beta^{o,a} = \frac{\mathcal{P}\left(o|s,a\right)\beta\left(s\right)}{\Sigma_{p' \in \mathcal{P}_o} \mathcal{P}\left(o|s',a\right)\beta\left(s'\right)} \tag{2}$$

From Beliefs to Decisions



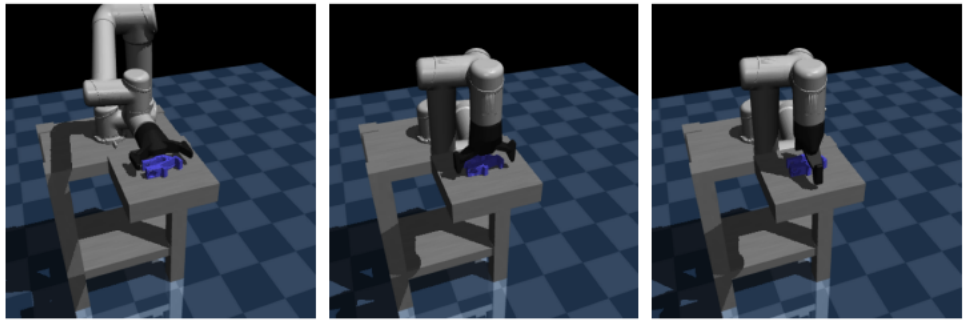


From Beliefs to Decisions



(a) (b) (c)

Fig. 2: Example stable poses.



(a) (b) (c)

Fig. 3: Example grasp configurations from which the actor can attempt to grasp the object - each configuration is associated with a score, i.e., probability of success.

From Beliefs to Decisions

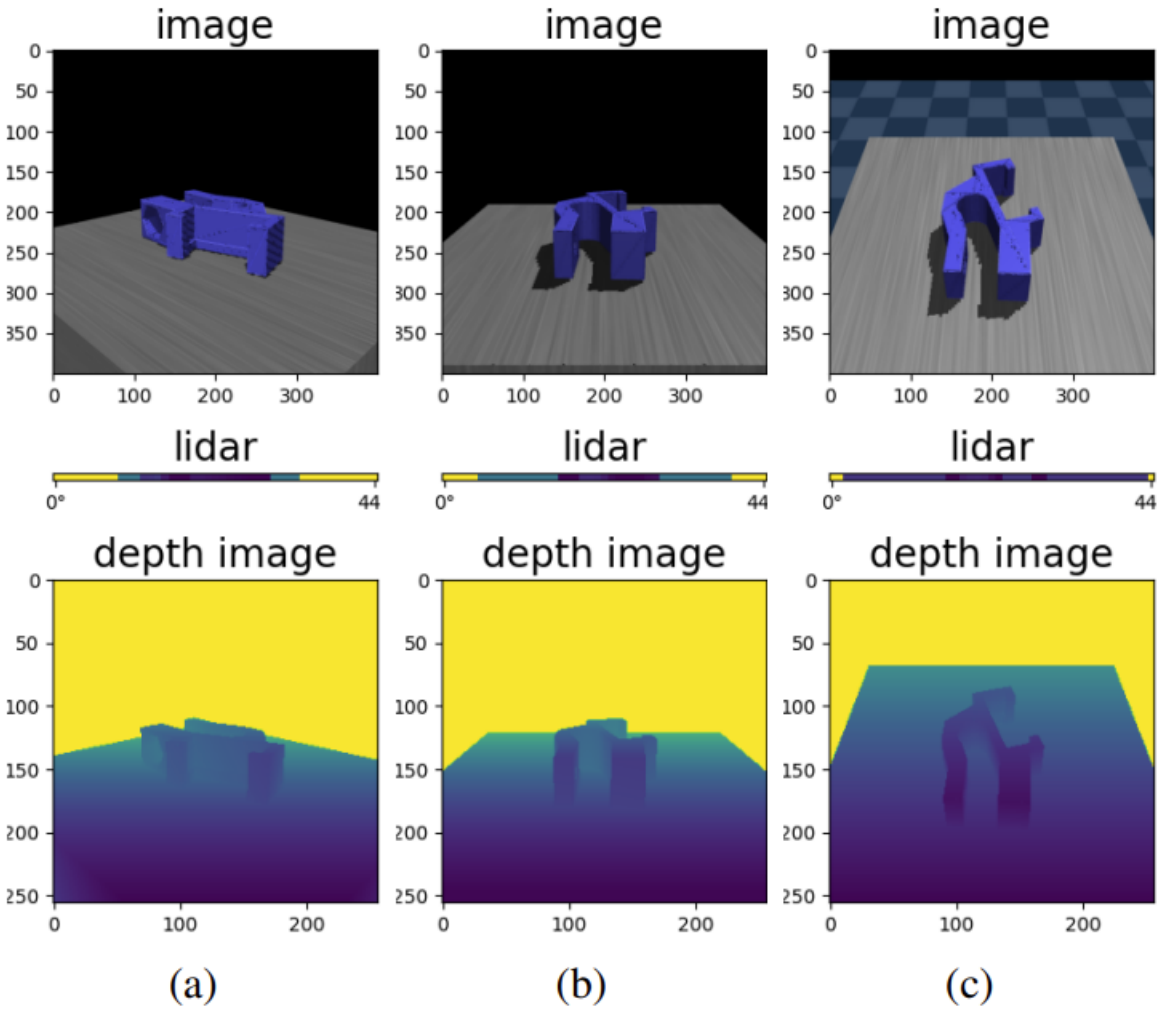


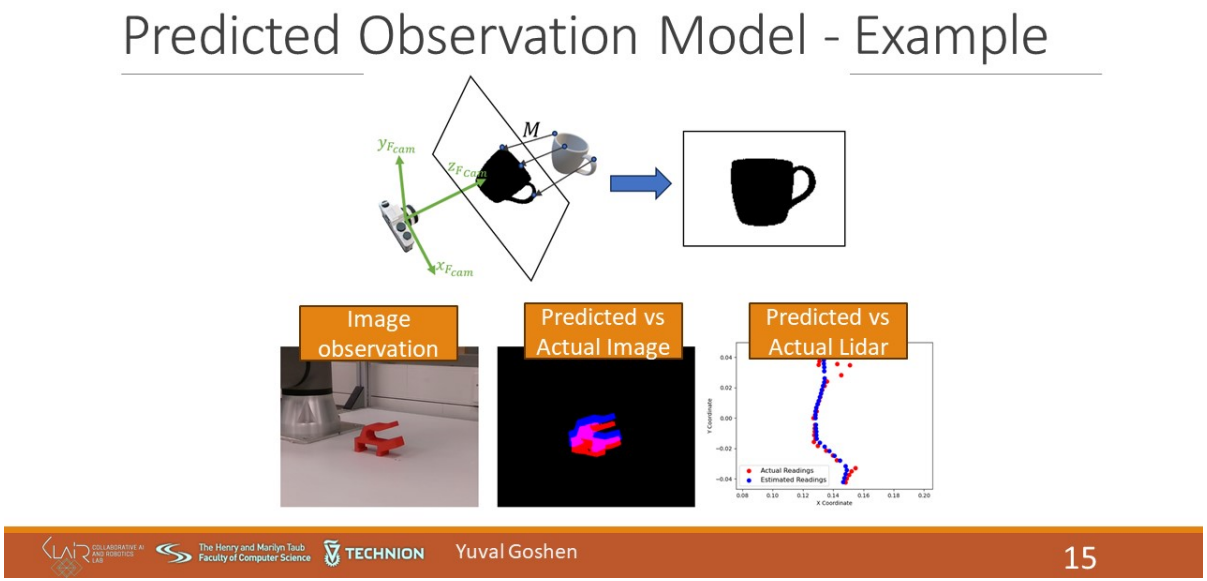
Fig. 4: Example sensor configurations and corresponding observations for a given stable pose. Each column represents the RGB image [top] lidar reading [middle] and depth image

Definition (Sensor Function)

Given object pose $p \in \mathcal{P}_o$ and sensor configuration $q \in \mathcal{Q}$, sensor function $\mathcal{O} : \mathcal{P}_o \times \mathcal{Q} \mapsto \Omega$ is a random function, such that if $o = \mathcal{O}(p, q)$ then $P(o|p, q)$ provides the conditional probability of obtaining the observation o when the sensor configuration is q and the object pose is p.

Typically, the actual distribution is not known, and we use a *predicted sensor function* $\tilde{\alpha}$ and a *predicted observation probability* $\hat{P}$, which may be incorrect or inaccurate.

Challenge: the probability of an observation is near 0.



From Beliefs to Decisions

We use a similarity score, $\omega : \Omega \times \Omega \mapsto [0, 1]$ to compare the predicted and received observations:

$$\hat{P}(o|p, q) = \frac{\omega(\tilde{\alpha}(p, q), o)}{\int_{p' \in \mathcal{P}_o} \omega(\tilde{\alpha}(p', q), o) dp'} \tag{3}$$

The literature is rich of various definitions for ω, which may vary between applications and sensor types.

From Beliefs to Decisions

For **belief update** we use a Bayesian filter such that for any observation $o \in \Omega$ taken from sensor configuration $q \in \mathcal{Q}$, the updated pose belief $\beta^{o,q}(p)$ for pose $p \in \mathcal{P}_o$ is given as

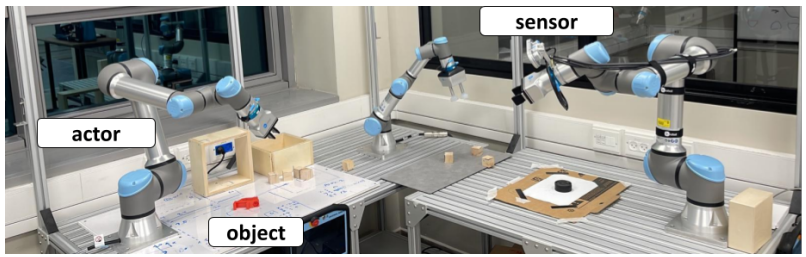
$$\beta^{o,q}(p) = \frac{\hat{P}(o|p,q)\,\beta(p)}{\int_{p' \in \mathcal{P}_o} \hat{P}(o|p',q)\,\beta(p')\,dp'} \tag{4}$$

where $\beta(p)$ is the estimated probability that p is the object pose prior to considering the new observation o.

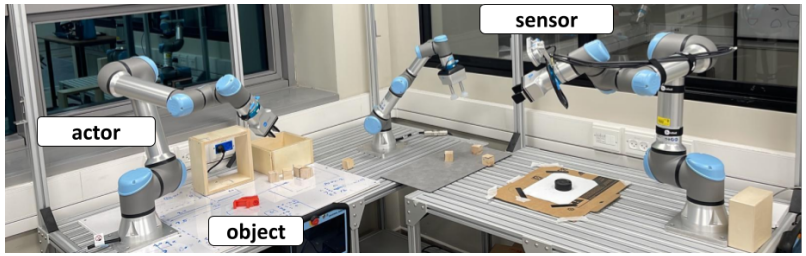
Value of Assistance (VOA) for Grasping

Given the actor's belief $\beta_a \in \mathcal{B}$, the belief of the helping agent $\beta_h \in \mathcal{B}$, the predicated observation probability $\hat{P}$, sensor configuration $q \in \mathcal{Q}$, and the actor's belief update function τ_a,

$$U_\alpha^{VOA}(\beta_h, \beta_{ac}) \stackrel{\text{def}}{=\!=} \tag{5}$$
$$\mathbb{E}_{p\sim\beta_h}\left[\mathbb{E}_{o\sim\hat{P}(o|p,q)}\left[\gamma(q_g^*(\beta_a^{o,q}), p)\right] - \gamma(q_g^*(\beta_{ac}), p))\right].$$



From Beliefs to Decisions





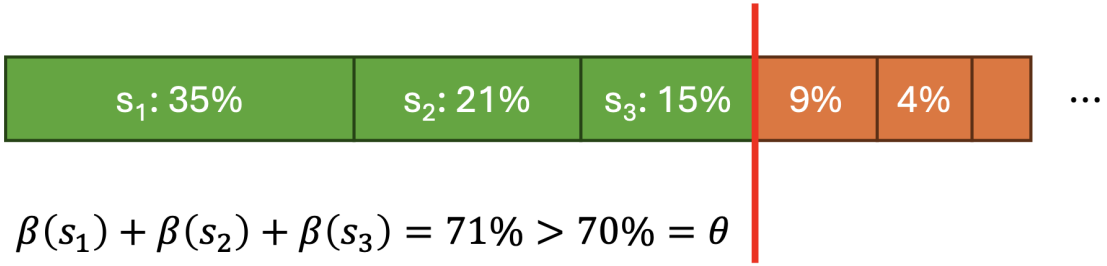
(a) P1 (b) P2 (c) P3 (d) P4 (e) P5 (f) P6

Is this a good observation?

Minimal Belief Sets

Don't use perception best-guess. Plan for multiple states. How many do we need?

$b = \{s_1, s_2, s_3\}$



$\beta(s_1) + \beta(s_2) + \beta(s_3) = 71\% > 70\% = \theta$

Probability-Ranked Valid State Search

$$Z = \sum_{s \vDash \mathcal{C}} \tilde{\beta}(s) = 1 - (\tilde{\beta}(110) + \tilde{\beta}(111)) = 0.72$$

Input

$\beta^F = [0.4, 0.7, 0.9]$
$\theta = 0.8$
$\mathcal{C} =$ first two features are mutually exclusive

\# - order of extraction from Q

1 011 $\tilde{\beta} = 0.6 \cdot 0.7 \cdot 0.9 = 0.378$, $\beta^{\mathcal{C}} = \frac{\tilde{\beta}}{Z} = 0.525$

2 111 $\tilde{\beta} = 0.4 \cdot 0.7 \cdot 0.9 = 0.252$, $\beta^{\mathcal{C}} = 0$ (Violates constraints)

3 001 $\tilde{\beta} = 0.162$, $\beta^{\mathcal{C}} = \frac{\tilde{\beta}}{Z} = 0.225$

010 $\tilde{\beta} = 0.042$, $\beta^{\mathcal{C}} = \frac{\tilde{\beta}}{Z} \approx 0.058$

drop visited

011

4 101

110

101

011

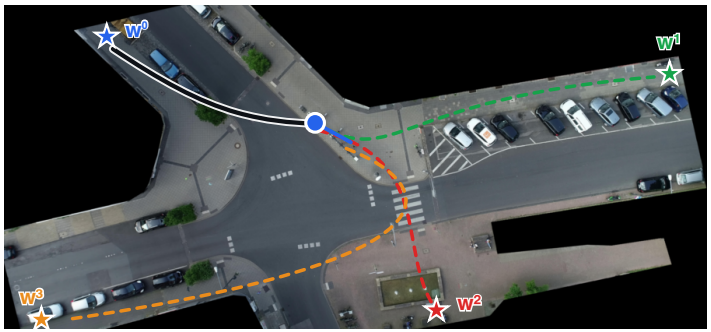
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Goal Recognition under Uncertainty

Watching a robot move: *where is it going?*

Belief over goals:

$$P(g \mid o_{1:t}) \;\propto\; p(o_{1:t} \mid g)\, P(g)$$



- Same Bayesian update as the filter, but the state is the goal
- Likelihood from a learned diffusion prior over trajectories, with no hand-written dynamics model
- Keeps multimodal, calibrated beliefs on ambiguous prefixes

